Notes on c=1 MQM and entanglement entropy

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1 Review of c=1 MQM

1.1 Continuum theory

From the sigma model to the gauge fixed action. We consider a free scalar coupled to two-dimensional Euclidean gravity, with a cosmological constant and a topological Einstein term on a smooth, compact two dimensional manifold $\mathcal{M}_h$ with genus h and metric g.

$$\mathcal{S}_\sigma^{(h,g,\Phi_0)} = \frac{1}{4\pi}\int_{\mathcal{M}_h} d^2x\sqrt{g}\left(g^{\mu\nu}\partial_\mu X\partial_\nu X + \Phi_0\mathcal{R} + 4\pi\mu\right) . \tag{1.1}$$

Φ_0 is a constant and as we will see later is connected to the constant part of the linear dilaton background. The Gauss-Bonnet theorem states that $\frac{1}{4\pi}\int d^2x\sqrt{g}$ is a topological number known as the Euler characteristic $\chi_h = 2-2h$. The partition function is defined through the path-integral over all possible metrics on $\mathcal{M}_h$ modulo Diffeomorphism and the boson X

$$\mathcal{Z} = \sum_{h\geq 0}(g_0)^{-\chi_h}\int \frac{[Dg_{\mu\nu}^{(h)}]}{\mathrm{Vol(Diffeo)}}[DX]e^{-\mathcal{S}_\sigma^{(h,g)}} , \tag{1.2}$$

where we introduced the notation $g_0 = e^{\Phi_0}$. Note that we remove the Φ_0 from $\mathcal{S}^{(\sigma,h,\Phi_0)}$ since we have extracted the topological dependent factor from the action. We can gauge fix the metric to the fiducial metric $g_{\mu\nu} = e^{2b\phi}\hat{g}_{\mu\nu}^{(\tau)}$. The price to pay is the introduction of b, c ghost fields. The measure is not invariant under the previous map. Also, the cosmological constant term yields a non trivial ϕ dependence (because of the cc term, even classically Weyl symmetry is broken). Combined together, they yield the Liouville action

$$\mathcal{S}_L^{(\hat{g}_h)} = \frac{1}{4\pi}\int d^2x\sqrt{\hat{g}_h}\left(\hat{g}_h^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + Q\hat{\mathcal{R}}\phi + 4\pi\mu e^{2b\phi}\right) . \tag{1.3}$$

We emphasize that the term in front of the Ricci scalar term comes from the anomaly of the measure. The remaining intergal to perform is an integral over the conformal mode ϕ known as the Liouville field and an integral over the moduli space parametrized by τ introduced previously.

$$M_h = \frac{\text{Metrics on } \mathcal{M}_h}{\text{Diffeo}\times\text{Weyl}} . \tag{1.4}$$

Hence, if modular invariance is satisfied, the partition function finally reads

$$\mathcal{Z} = \sum_{h\geq 0}(g_0)^{-\chi_h}\int d\tau\int[D\phi][DX][Db][Dc]e^{-(\mathcal{S}_\sigma+\mathcal{S}_{b,c}+\mathcal{S}_L)} , \tag{1.5}$$

where V_h is the volume of the moduli space of M_h. Let's focus on the Liouville sector for a moment. We can exploit that ϕ is integrated over to extract the μ dependence of the Liovuille partition function. In fact, assuming the measure is invariant under $\phi\to\phi-\frac{\log(\mu)}{2b}$, one obtains

$$\mathcal{Z}_L^{(h)}(\mu) = \mu^{-\frac{Q\chi_h}{2b}}\mathcal{Z}_L^{(h)}(\mu=1) . \tag{1.6}$$

Liouville action and Target space interpretation The different terms appearing in the Liouville+matter action have a meaning in terms of Target space fields. Let's recall the general form of the generalized non linear sigma model in presence of a dilaton $\Phi(Y)$, a Tachyon condensate $T(Y)$ and non trivial metric $G_{\mu\nu}(Y)$.

$$\mathcal{S} = \frac{1}{4\pi}\int d^2x\sqrt{\hat{g}}\left(T(Y) + Q\hat{\mathcal{R}}\Phi(Y) + g^{ab}G_{\mu\nu}\partial_a Y^\mu \partial_b Y^\nu\right) . \tag{1.7}$$

We see by comparison to the Liouville + matter action that $Y = (X, \Phi)$ and

$$T = \mu e^{2b\phi} \quad , \quad G_{\mu\nu} = \delta_{\mu\nu} \quad , \quad \Phi = \frac{\Phi_0}{Q} + \phi\,. \tag{1.8}$$

This is important to note that the background (1.8) is solution of the one-loop vanishing of the β functions only for $\mu = 0$. For $\mu \neq 0$ the Liouville interaction should be interpreted as a tachyon-wall background rather than as a naive linear-dilaton solution of the free beta-function equations.

There is also an easy way that the potential plays the role of a Tachyon condensate. Indeed, the vertex operator reads

$$\mathcal{V}_T = e^{ik.Y} := e^{ikX+2\alpha\phi}\,. \tag{1.9}$$

The marginality condition fixes $\Delta_X + \Delta_\phi = \Delta_X + \alpha(Q-\alpha) = 1$. Solving the previous condition with $\Delta_X = 0$ and selecting the state satisfying the Seiberg bound one obtains $\alpha = b$. We denote the integrated vertex operator as

$$V_T = \int d^2x\sqrt{\hat{g}}\,\mathcal{V}_T\,. \tag{1.10}$$

The Tachyon condensate results in an insertion of the infinite sum of $(\mu\, V_T)^n$

$$\sum_{n=0}^{\infty}\frac{\mu^n}{n!}\int d^2x_1\sqrt{\hat{g}}...\int d^2x_n\sqrt{\hat{g}}\,e^{2b\phi(x_1)}...e^{2b\phi(x_n)}\,. \tag{1.11}$$

Remark that we effectively started with a sigma model with one boson and landed on a theory which has now two bosons X, ϕ. We say that from the non-decoupling of the conformal mode, a new dimension has emerged. This describes a inhomogeneous 2D space (ϕ, X). Such inhomogeneity emerges in the ϕ direction due to the Liouville potential. The string coupling depends on the value of the dilaton and thus on the position in the ϕ direction as the dilaton and the Liouville conformal mode are related by (1.8). An observer sitting on the Liouville wall defined by the condition

$$\mu e^{2b\phi_\star} = 1 \Rightarrow \Phi_\star = \frac{\Phi_0}{Q} + \phi_\star = \Phi_0 - \frac{\log(\mu)}{2b}\,. \tag{1.12}$$

observes the scattering of states with the string coupling $g_\star = e^{Q\Phi_\star}$. The amplitude of the process should be perturbatively described by a genus expansion in the string coupling at the position of the Liouville wall and this is exactly what the KPZ scaling relation gives (see (1.5), (1.6)).

Genus 0 Lagrangian On the sphere $h = 0$, we choose a representant of the conformal class $\hat{g}_h$ such that the metric is flat on the complex plane $(z, \bar{z})$ except at one point which is taken at infininity. If one writes $ds^2 = 2dzd\bar{z}$, then the Liouville Lagrangian density on the sphere is

$$\begin{aligned} L_{\text{bulk.}} &= \frac{1}{4\pi}\left(\partial_z\phi\partial_{\bar{z}}\phi + Q\hat{\mathcal{R}}\phi + \mu e^{2b\phi}\right) \\ &= \frac{1}{4\pi}\left(-\phi\partial_z\partial_{\bar{z}}\phi + Q\hat{\mathcal{R}}\phi + \mu e^{2b\phi} + \text{total derivative}\right)\,. \end{aligned} \tag{1.13}$$

Choosing the representative of the conformal class which is flat on $\mathbb{C}$ trades the bulk curvature coupling for an asymptotic boundary condition at $z = \infty$. Thus the curvature term does not appear

explicitly in the local Lagrangian density on the finite plane, but its effect is retained through the background-charge condition on ϕ at infinity.

$$\phi(z,\bar{z}) \underset{z\to\infty}{=} -2Q\log(z\bar{z}) + \mathcal{O}(1)\,. \tag{1.14}$$

In fact, $\partial_z\partial_{\bar{z}}\log(z\bar{z}) = 2\pi\delta^2(z,\bar{z})$.

Adding a boundary to the manifold Consider now the possibility of adding a boundary and a non trivial interaction.

$$\mathcal{S}_L^{\text{bound.}} = \frac{1}{4\pi}\int d^2x\sqrt{\hat{g}_h}\left(\hat{g}_h^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + Q\hat{\mathcal{R}}\phi + 4\pi\mu e^{2b\phi}\right) + \frac{1}{2\pi}\int_{\partial M} ds\sqrt{\hat{h}}\hat{K}\phi + \mu_b\int_{\partial M} ds\sqrt{\hat{h}}e^{b\phi}\,. \tag{1.15}$$

h is the physical metric at the boundary and is related to the fiducial metric via $\sqrt{h} = e^{b\phi}\sqrt{\hat{h}}$ and $\hat{K}$ is the extrinsic curvature at the boundary. μ_b is the boundary cosmological constant but should also be understood as the tension/charge of the associated brane. The variational principle is well defined as long as it satisfies Dirichlet boundary conditions at the boundary. It can also satisfies Neumann boundary conditions if

$$n^\mu\partial_\mu\phi + \hat{K} + 2\pi b\mu_b = 0\,. \tag{1.16}$$

A Boundary states: Ishibashi states vs Cardy states

We first introduce Virasoro primary states. Highest-weight representations of the Virasoro algebra are generated from states $|V_P\rangle$ satisfying

$$L_0\,|V_P\rangle = \Delta_P\,|V_P\rangle\,, \qquad L_n\,|V_P\rangle = 0 \quad (n>0), \qquad \Delta_P = \frac{Q^2}{4} - P^2. \tag{A.1}$$

We do not impose any normalization condition on the states $|V_P\rangle$.

On a conformal representative of a manifold with boundary, preserving one copy of the conformal symmetry requires the absence of momentum flow through the boundary,

$$n_\mu T^{\mu\nu} = 0. \tag{A.2}$$

At the quantum level, this condition translates into the gluing condition

$$\left(L_n - \bar{L}_{-n}\right)|B\rangle = 0, \qquad n\in\mathbb{Z}, \tag{A.3}$$

where the boundary state $|B\rangle$ is viewed as a closed-string state.

For each Virasoro representation generated by a primary state $|V_P\rangle$, there exists an Ishibashi state, denoted by $|P\rangle$, which solves the gluing condition. It can be written formally as

$$|P\rangle = \sum_{N\geq 0}\sum_{I,J\in\mathcal{P}_N}\left(S^{(N)}\right)^{-1}_{IJ} L_{-I}\,|V_P\rangle \otimes \overline{L_{-J}\,|V_P\rangle}, \tag{A.4}$$

where $\mathcal{P}_N$ denotes the set of partitions of the level N, $L_{-I} = L_{-i_1}\cdots L_{-i_k}$ for $I=(i_1,\ldots,i_k)$ with $\sum_a i_a = N$, and $S^{(N)}$ is the Shapovalov matrix of Virasoro descendants at level N. In general, physical boundary states also called Cardy states need not to be Ishibashi states. Physical boundary states $|s_1\rangle$, $|s_2\rangle$ satisfy the open/close string duality Cardy condition. Let an annulus of unit width for which the circle has length β. Then

$$\begin{aligned} Z_{s_1\,s_2}(\beta) &= \mathrm{Tr}_{\mathcal{H}_{\text{open}}}[\mathrm{e}^{-\beta \mathrm{H}_{s_1\,s_2}}] = \sum_{\mathrm{h}} \mathrm{n}_{\mathrm{h}}^{s_1\,s_2}\chi_{\mathrm{h}}(\beta) \\ &= \langle s_1|\,e^{-H/\beta}\,|s_2\rangle\ . \end{aligned} \tag{A.5}$$

This condition should be understood as determining $n_h^{s_1\,s_2}$.